

# **The Anatomy of a Regression Model**

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## 1 Introduction

Fitting a regression model comes with a whole set of decisions based on how the model is “built”, or specified. This lecture will provide you with a framework of what these decisions are, and how they correspond to components of the model. These include things like what the outcome and how it is coded, the right hand side and how its variables enter, the link function, the distribution and what that does and does not give you, and (if so desired) the offset of the model, why it’s there and what it does. Each of these decisions comes with its own consequences for what the fitted numbers mean once you fit the model.

In this lecture, we write down some models and discuss the implications and what each piece of the model means. We then look at these pieces in turn: the left hand side, the right hand side, the split between target and nuisance functions, the link function, the distribution, and the offset. Think of this as a lesson on the “anatomy” of a regression model. We close with a discussion of model flexibility: parametric, semiparametric, and nonparametric models, and what this means.

**What to read:** Clark & Berry. Estimation. Chapter 6 of *Models Demystified*: [m-clark.github.io/book-of-models](https://m-clark.github.io/book-of-models)



### **Context Note:** Regression Anatomy and the Frisch-Waugh-Lovell Theorem

Note that my use of the “anatomy” metaphor here might lead to some confusion. In econometrics, “regression anatomy” refers to a specific algebraic result, popularized using that phrase by [Angrist and Pischke \(2009\)](#): the coefficient on  $X_1$  in a multiple regression equals the coefficient from a simple regression of  $Y$  on the residual of  $X_1$  after  $X_1$  is itself regressed on every other right hand side term ([Frisch and Waugh, 1933](#); [Lovell, 1963](#)). That is a theorem about partialling out, and serves as the theoretical foundation of methods like residual-on-residual regression ([Robinson, 1988](#); [Naimi et al., 2026](#)) and some versions of double-debiased machine learning ([Chernozhukov et al., 2018](#)). This week, “anatomy” means something plainer: the structural dissection of the model object, its slots and the choices they encode. The two senses are cousins rather than strangers. The partialling result is the algebra behind a fact we use repeatedly below, that every coefficient in a fitted model is conditional on everything else on the right hand side. We use the fact and leave the theorem to the references.

## 2 A Data Example

It will be a great help if we could keep our explanations here concrete. To do this, we have to not only pick a dataset, but also a general question that helps

us define what we want to use our regression model for. This creates a bit of a problem for us. If we are focusing on a descriptive, predictive, or causal analysis, we might make different choices about how we specify things in our regression model (which is what this lecture will be about).

However, if we don't pick a specific question, we lose the ability to connect our decisions to their justifications. So, to avoid losing this, we'll work with the National Health and Nutrition Examination Survey Epidemiologic Follow-up Study, or NHEFS data ([Hernán and Robins, 2020](#)), included in the course repository, and seek to answer the question about how much quitting smoking between 1971 and 1982 affects weight change between those two time points. The NHEFS enrolled adults aged 25 to 74, measured them in 1971, measured them again in 1982, enabling us to try to answer this question:

```
nhefs <- read_csv(here("data", "nhefs.csv")) %>%
  select(qsmk, wt82_71, wt82, wt71, exercise, sex, age, race,
         marital, smokeintensity, smkintensity82_71,
         death, yrdth) %>%
  filter(!is.na(wt82_71))

dim(nhefs)
```

```
## [1] 1566 13
```

The dataset contains 1,566 people with weight measured at both visits. The variables we have at our disposal in the NHEFS data include: `wt82_71` (weight change in kg between the visits), `wt82` (weight in 1982), and `wt71` (weight in 1971), the first of which will be our outcome; `qsmk` (quit smoking between the visits, 1 = yes, which is the exposure of interest); and `age`, `sex`, `race`, `exercise`, and `marital`, `smokeintensity` (cigarettes per day in 1971), which will serve as our confounders.

### 3 A Regression Model

A general regression model can be written as:

$$g[E(Y | X)] = \underbrace{\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p}_{\text{r.h.s}}$$

The left hand side of this equation contains the conditional expectation of the outcome,  $E(Y | X)$ , and a link function  $g$  that determines the scale on which the right hand side is additive. We'll cover link functions soon; the simplest case is the "identity" link, which means there is no function, so  $g[E(Y | X)] = E(Y | X)$ . Sometimes, we write the conditional mean as  $\mu$ , yielding  $g(\mu) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$ , and we can make the link explicit by writing  $E(Y | X) = g^{-1}(\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p)$ .

The right hand side can be simplified using matrix notation as:

$$g[E(Y | X)] = X\beta$$

In this notation,  $\beta$  is a  $p \times 1$  vector of regression coefficients, and  $X$  is an  $n \times p$  matrix, with a column of 1's to incorporate the intercept parameter  $\beta_0$ . This  $X$  matrix is often referred to as the **design matrix**. Note that bold symbol font in regression is conventionally used for vectors or matrices, but not always. In fact, often, researchers will simply write:  $X\beta$ , which will be the convention we follow here.<sup>1</sup>

A few pieces of a regression model are missing from this equation, sometimes present explicitly, sometimes not. The first is the distribution, our model for how  $Y$  varies around its conditional mean. This is often required in the context of generalized linear models, and provides the framework we need to estimate parameters via MLE, as we saw in Week 4. The second is the assumption that the observations are independent of one another, which never appears in the equation at all; this is usually essential (in some way), and we'll see it come up later in the course. Finally, a model may include an offset, which is a variable on the right hand side of the model whose coefficient is set to one, and not estimated. We'll see why we might want such a thing in its own section below.

Next, we focus in on each of these pieces to discuss details that are important for us to consider as we proceed with regression in our applied work.

<sup>1</sup> Note another matrix convention which is rarely deployed is to denote the dimensions of the covariate vector and design matrix directly within the regression function:

$$g[E(Y | X)] = \sum_{n \times p} \sum_{p \times 1} \beta$$

This is not always important to do, but can be useful when evaluating the model from a matrix multiplication standpoint (e.g., say you're writing code using matrix multiplication to get things to work together, see Section 3 of the math foundation notes [here](#)).

## 4 The Left-Hand Side

Making decisions about the left hand side of a regression model is one component of how we specify (and misspecify) the model, and shapes the scale of the descriptimand, predictimand, or estimand: basically, it selects the scale of the outcome summary that the model will estimate (Week 2's discipline of selecting the occurrence measure is here as a modeling decision).

To see this operate, let's use the NHEFS weight change outcome on the left hand side. The simplest thing we can do with regression is specify a simple mean. This is the null model,  $E(Y) = \beta_0$ , which has one parameter, the intercept, which is the population mean. In fact, estimating a mean is regression (i.e., regressing an outcome on an intercept term):

```
mean(nhefs$wt82_71)
```

```
## [1] 2.6383
```

```
summary(lm(wt82_71 ~ 1, data = nehs))$coefficients[,1:2]
```

```
##      Estimate Std. Error
## 2.6382998   0.1991249
```

We can add a single indicator term to this model, for instance, whether an individual in the data quit smoking or not:

$$E(Y | X) = \beta_0 + \beta_1 X$$

and with  $X$  coded 0/1,  $\beta_1$  is the difference in the mean of  $Y$  among those with  $X = 1$  versus  $X = 0$ , which here compares the **mean** weight change between quitters and non quitters:

```
round(summary(lm(wt82_71 ~ qsmk, data = nehs))$coefficients, 2)
```

```
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    1.98         0.23    8.67      0
## qsmk           2.54         0.45    5.63      0
```

Quitters gained, on average, 2.54 kg more than non quitters over the eleven years. Note here that coding matters (we'll return to this issue again). With a  $k$  category grouping variable,  $k - 1$  indicators encode every subgroup contrast when an intercept is included in the model. We can drop the intercept and read off the subgroup means directly.

```
summary(lm(wt82_71 ~ -1 + factor(qsmk), data = nhfs))$coefficients
```

```
##              Estimate Std. Error  t value    Pr(>|t|)
## factor(qsmk)0  1.984498  0.2288285  8.672423 1.040008e-17
## factor(qsmk)1  4.525079  0.3887295 11.640690 4.281980e-30
```

```
nhfs |>
  group_by(qsmk) |>
  summarise(mean_wt_change = mean(wt82_71))
```

```
## # A tibble: 2 x 2
##   qsmk mean_wt_change
##   <dbl>         <dbl>
## 1     0             1.98
## 2     1             4.53
```

This is a sense in which regression unifies elementary statistics: the comparisons of means and proportions from a first course in statistics are really all regressions, and the machinery scales from there without really changing the character of what is happening in the regression model (i.e., elementary statistical comparisons become generalized to the multivariate case) ([Carlin and Moreno-Betancur, 2025](#)).

The mean is one, albeit the most common choice, of summary measure for an outcome. Instead, we could use quantile regression models to compare a quantile of the same outcome distribution among the exposed versus unexposed group ([Koenker and Bassett, 1978](#)). For instance, if we let  $\tau = 0.5$  be the median, we can compare the median among those who quit and those who didn't using quantile regression:

$$F_{\tau}^{-1}(Y | X) = Q_{\tau}(Y | X) = \alpha_0(\tau) + \alpha_1(\tau)X$$

Note the difference is not in the outcome or the independent variable, or really anything pertaining to the data. What's different is the function we used to operationalize the left hand side of the regression equation. This tells a slightly different story in the NHEFS data:

*## to run this, you'll need the quantreg package installed and loaded*

```
round(summary(rq(wt82_71 ~ qsmk, tau = 0.5, data = nhefs))$coefficients, 2)
```

| ##             | Value | Std. Error | t value | Pr(> t ) |
|----------------|-------|------------|---------|----------|
| ## (Intercept) | 2.15  | 0.20       | 10.80   | 0        |
| ## qsmk        | 1.82  | 0.45       | 4.03    | 0        |

The difference in **median** weight change is 1.82 kg, even though we use the same data, same right hand side, different left hand side target, different number. Neither is wrong, they just answer different questions, and a distribution can easily shift more in its mean than in its middle when one tail moves.

We can visualize this by plotting the empirical cumulative distribution function of weight change separately for quitters and non quitters. Then the  $\tau$  quantile of each group is the point where its curve crosses a horizontal line drawn at height  $\tau$ . A quantile regression with a single binary exposure estimates the horizontal distance between the two curves at that height:

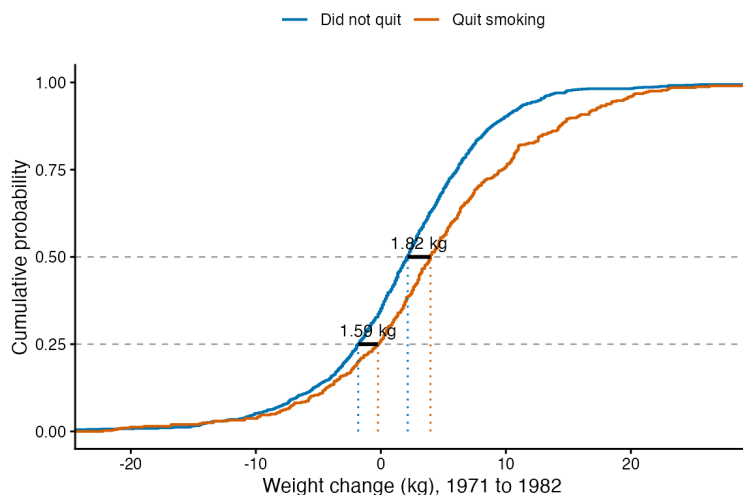


Figure 1: Empirical cumulative distribution functions of weight change by quitting status. The  $\tau$  quantile of each group is the point where its curve crosses the horizontal line at height  $\tau$ ; the dotted drops mark these quantiles. Quantile regression with a binary exposure estimates the horizontal distance between the two curves at that height: 1.59 kg at the 25th percentile and 1.82 kg at the median, the coefficients from the models in this section.

The figure also shows why the mean and median differences disagree: the gap between the two curves widens as we move up the distribution, which is

exactly the upper tail shifting more than the middle. Note, we can also express interest in another quantile, say the 25th percentile instead of the median:

```
round(coef(rq(wt82_71 ~ qsmk, tau = c(0.25, .5), data = nhefs)), 2)
```

```
##           tau= 0.25 tau= 0.50
## (Intercept)   -1.81    2.15
## qsmk          1.59    1.82
```

A third option is to dichotomize the outcome here. We can create an indicator of greater than median weight change, and the median here is 2.6 kg. Before adopting any such cutpoint, we should ask if the threshold is meaningful? If the scientific interest is in weight gain specifically, and the median in our dataset were negative, the median would be a poor cutpoint because it would include those who also experienced weight loss. Is the threshold stable? The NHEFS median is a property of this sample, but another cohort's median can be different. Results obtained using variables defined on the basis of sample summaries are often not generalizable.

```
round(summary(glm(as.numeric(wt82_71 > median(wt82_71)) ~ qsmk,
                  data = nhefs,
                  family = binomial(link = "log")))$coefficients, 2)
```

```
##           Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -0.76      0.03  -24.31      0
## qsmk          0.24      0.05   4.71      0
```



**Note:** The Mean of a Transformation Is Not the Transformation of the Mean

A common instinct with a skewed continuous outcome is to log transform it and fit a linear model to  $\log Y$ . That model's left hand side is  $E(\log Y \mid X)$ . A generalized linear model with a log link has the left hand side  $\log E(Y \mid X)$ . These are different quantities: by Jensen's inequality,  $E(\log Y) \leq \log E(Y)$ , with equality only when  $Y$  is constant. Exponentiating a coefficient from the log outcome model therefore does not return a ratio of means. It returns a ratio of geometric means, and naively back transforming as if it were a ratio of arithmetic means can yield numbers that are wrong.

The note is easy to verify. Take weight in 1982 (a positive variable, so we can actually take the log) as the outcome and compare three fits: a linear

model for the mean, a linear model for the log of the outcome, and a log link model for the mean itself:

```
round(coef(lm(wt82 ~ qsmk, data = nhfs)), 3)
```

```
## (Intercept)      qsmk
##      72.287      4.593
```

```
round(coef(lm(log(wt82) ~ qsmk, data = nhfs)), 4)
```

```
## (Intercept)      qsmk
##      4.2579      0.0606
```

```
round(coef(glm(wt82 ~ qsmk, data = nhfs,
               family = gaussian("log"))), 4)
```

```
## (Intercept)      qsmk
##      4.2806      0.0616
```

The first model reports a 4.59 kg higher mean weight among quitters. Exponentiating the second model's coefficient gives  $\exp(0.0606) = 1.0625$ , the ratio of geometric mean weights. Exponentiating the third gives  $\exp(0.0616) = 1.0635$ , the ratio of arithmetic means, and indeed the two arithmetic means are 76.88 and 72.29 kg, whose ratio is 1.0635. The gap between 1.0625 and 1.0635 is small in these data because weight is not severely skewed, but the two numbers are answers to different questions, and with a heavier tail in the outcome's distribution, the gap grows.

## 5 The Right-Hand Side

The right hand side is where most of a model's choices accumulate. It's important to understand what the software builds from what you actually put into the right hand side of a regression model. A model formula is a recipe for a design matrix. In R, one way to construct a design matrix manually is to use the `model.matrix()` function<sup>2</sup>:

<sup>2</sup> Note that you often don't have to construct the design matrix manually. Typically, you put in the variables you want on the right hand side of the regression equation, and R builds a design matrix automatically (sometimes, using `model.matrix()` internally)

```
head(model.matrix(~ qsmk + factor(exercise), data = nhfs), 4)
```

```
##      (Intercept) qsmk factor(exercise)1 factor(exercise)2
## 1             1    0                    0                    1
## 2             1    0                    0                    0
## 3             1    0                    0                    1
## 4             1    0                    0                    1
```

Three features deserve attention. The column of 1's is the intercept term. The `qsmk` column is the variable as supplied. And the three level exercise variable has become two indicator columns, with level 0 absent: R chose it as the **referent**, and every exercise coefficient will be a contrast against it (this is a feature of `factor()`). The referent choice is arbitrary here (by default, alphabetical or numerical rather than scientific). However, whether this actually matters depends on a few things, notably the distinction between whether the variable you're coding is in the target or nuisance function component of the model (a concept we'll see in a moment).

Next we consider units. Consider the association between change in smoking intensity (cigarettes per day) and weight change:

```
round(summary(lm(wt82_71 ~ smkintensity82_71,
                 data = nhfs))$coefficients[2, 1:2], 3)
```

```
##      Estimate Std. Error
##      -0.044      0.015
```

Each additional cigarette per day is associated with a 0.044 kg lower weight change. This estimate is very small. This is because the scientific question here coded in this regression model is denominated in single cigarettes. A one unit change that is substantively trivial can make a potentially important association look negligible, the same way a 1 mm Hg difference in blood pressure or a one minute difference in daily exercise would. Rescaling can be done in a number of ways, for example:

```
round(summary(lm(wt82_71 ~ I(smkindensity82_71/5),
                  data = nhefs))$coefficients[2, 1:2], 3)
```

```
##      Estimate Std. Error
##      -0.222      0.073
```

Now the coefficient, 0.22 kg per five cigarettes per day, sits on a scale a reader can weigh.<sup>3</sup> The general recipe: to interpret a coefficient per  $a$  units, enter the variable as  $I(x/a)$ .

A tempting alternative is to standardize the exposure into a z score,  $(X - \bar{X})/SD(X)$ , so the coefficient reads “per standard deviation.” But you should resist the temptation for variables of scientific interest (Greenland et al., 1991). The standard deviation is a property of the study, not of nature: it changes with eligibility criteria, with the sampling design, measurement error, and many other fairly arbitrary features of how the study was designed, so the same biological relationship yields different “per SD” coefficients in different studies, and the exposure’s scale ends up confounded with features of the design (Greenland et al., 1991). If two studies of smoking intensity report effects per standard deviation, and their standard deviations are 8 and 14 cigarettes, their coefficients are not comparable, and neither is interpretable without undoing the transformation.

What about categorizing a continuous variable? Statisticians have argued against the practice for decades, on efficiency grounds (Altman and Royston, 2006; Bennette and Vickers, 2012; Schellingerhout et al., 2009), and the argument is easy to demonstrate. Simulate a continuous exposure related to an outcome, then estimate the association twice, once with the exposure as supplied and once dichotomized at its median:

```
set.seed(123)
simres <- NULL
for(i in 1:1000){

  n <- 100
  x <- exp(rnorm(n))
  y <- 1 + x + rnorm(n, sd = 10)
```

<sup>3</sup> The  $I()$  function is R’s “as is” operator. Inside a formula, symbols like  $+$ ,  $*$ , and  $\sim$  are formula operators; wrapping an expression in  $I()$  makes R treat them as arithmetic instead.  $\text{lm}(y \sim I(x/5))$  divides  $x$  by 5;  $\text{lm}(y \sim x/5)$  does something else entirely, and in this case returns an error.

There are settings where standardizing has a rationale, such as gestational weight gain, which is strongly correlated with gestational age. Even there, the usual advice is to standardize against the standard deviation in a defined target population rather than whatever the sample happens to supply.

```

dat_ <- tibble(y, x, x_bin = as.numeric(x > median(x)))

simres <- rbind(simres,
               rbind(tidy(lm(y ~ x, data = dat_))[2, 1:5],
                     tidy(lm(y ~ x_bin, data = dat_))[2, 1:5]))
}

simres %>%
  group_by(term) %>%
  summarise(mean_se = mean(std.error))

## # A tibble: 2 x 2
##   term    mean_se
##   <chr>    <dbl>
## 1 x        0.534
## 2 x_bin    2.04

```

Dichotomizing multiplies the average standard error by nearly four in this setting: information about where each person sits within their half of the distribution is simply discarded. Does that settle the matter? No. The efficiency argument is a statistical consideration (i.e., on the grounds of estimator efficiency, since the same data is giving us an estimator with much higher error). However, statistical considerations should not always outrank scientific considerations. Sometimes the underlying relationship itself has a threshold that justifies the cut: the glycemic cutoffs that define diabetes sit where diabetes-specific retinopathy, essentially absent below a narrow glycemic range in pooled data on more than 44,000 people, begins to appear ([Colagiuri et al., 2011](#)). Other times, researchers or clinicians are simply interested in the dichotomized version even though no threshold exists: maternal glucose shows smoothly continuous relationships with adverse pregnancy outcomes ([Metzger et al., 2008](#)), yet diagnostic cutoffs for gestational diabetes were adopted anyway ([Metzger et al., 2010](#)), because the decision to treat is binary even when the risk gradient is not. Either way, one should not adopt a hard position that continuous variables should never be dichotomized ([Farrington and Loeber, 2000](#); [DeCoster et al., 2009](#)).

Product terms are another choice to make about variables on the right hand side. In week 4 we saw that the kidney model, having no product term between infection and age, was committed to a single infection **difference** at every age: regress weight change on quitting and a cubic in age, and the model is committed to one quitting difference at every age from 25 to 74:

```
constrained_model <- lm(wt82_71 ~ qsmk + age + I(age^2) + I(age^3),
                        data = nhefs)

round(summary(constrained_model)$coefficients["qsmk", 1:2], 2)

##      Estimate Std. Error
##          3.13      0.44
```

That single difference is 3.13 kg across all ages in the data. To avoid making this assumption, and letting the data guide us as to whether or not the difference should change as a function of age, we can include an interaction in the model:

```
# to see what the "*" does:
formula(terms(~ qsmk*(age + I(age^2) + I(age^3)), simplify = TRUE))

## ~qsmk + age + I(age^2) + I(age^3) + qsmk:age + qsmk:I(age^2) +
##      qsmk:I(age^3)

flex_model <- lm(wt82_71 ~ qsmk*(age + I(age^2) + I(age^3)),
                data = nhefs)

pred_diff <- function(a){
  predict(flex_model, newdata = data.frame(qsmk = 1, age = a)) -
  predict(flex_model, newdata = data.frame(qsmk = 0, age = a))
}

round(c(age_30 = pred_diff(30), age_50 = pred_diff(50),
       age_70 = pred_diff(70)), 2)
```

```
## age_30.1 age_50.1 age_70.1
##      2.31      4.29     -1.44
```

With product terms, the model may report a different quitting difference at every age, and here it does: the difference in weight gain among those who quit smoking versus those who didn't is about 2.3 kg at age 30, 4.3 kg at 50, and negative 1.4 kg at 70. So, note that the “constant difference” versus “age varying difference” model is a commitment we make. What a product term *means*, and in particular the fact that its meaning depends on the scale of the model itself. Additionally, whether one model is “better” than the other depends on a whole range of considerations that we will revisit again and again in the course.

One more fact ties this section together. Every coefficient in a fitted model is conditional on everything else on the right hand side. Add a term, remove a term, or recode a term, and in general every other coefficient changes, because each one now describes the outcome's relationship with its variable within levels of a different adjustment set. This is why “the coefficient for smoking” is an incomplete phrase; the complete phrase names the model it came from. The partialling algebra of the context note above is the formal version of this fact, and this topic is covered in section [2.4.1 Partial Derivatives](#) of the “Math Foundations” notes. We will see examples of where this distinction really matters when we look at conditionally adjusted versus marginally standardized models.

## 6 Target and Nuisance Functions

In week 2, we previewed a distinction that we can now state clearly. Consider a logistic model for an outcome  $Y$ , an exposure of interest  $X$ , and a set of confounders  $C$  required for identification:

$$\text{logit}[P(Y = 1 \mid X, C)] = \underbrace{\alpha_1 X}_{\text{target}} + \underbrace{\alpha_0 + \alpha'_2 C}_{\text{nuisance function } \eta(C)}$$

The **target** is the piece of the model the estimand needs, here the coefficient  $\alpha_1$ . The **nuisance function**  $\eta(C)$  is everything the model carries so the target is identified and estimable. The target function is coded for interpretation because it is what provides us with the answer to our scientific question of

interest: the units, referent, and form should make the estimand legible and precisely relevant to what we want to know. The nuisance function is coded for performance of the target function. We are not interested in the effect of the confounders on the outcome (otherwise, they would be in the target function), so we choose coding and operationalization that makes the target function estimate efficient, accurate, and stable.



**Context Note:** The Table 2 Fallacy, From the Model's Side

We saw the Table 2 fallacy as a reporting problem in week's 2 and 4: a manuscript's Table 2 presents every coefficient from an adjusted model, and readers interpret each as an effect ([Westreich and Greenland, 2013](#)). The anatomy of this week explains the mechanics. The confounders were selected, measured, and coded to identify the exposure effect (the target function). Nothing about that process addresses the confounders' own confounders because they are in the nuisance function; identification was never attempted for them, so their coefficients are typically a blend of whatever causal, confounded, and selection forces act on those variables. If a confounder's effect is scientifically interesting, it deserves its own analysis with its own identification argument and its own nuisance function.

Coding variables in the nuisance function does not require the same set of considerations we might make when coding variables in the target function. Consider the NHEFS marital status variable. Suppose smoking intensity is the outcome, sex is the exposure of interest, and age and marital status are confounders to be adjusted. Marital status contains the following numeric codes:

```
table(nhefs$marital)
```

```
##
##      2      3      4      5      6      8
## 1237   88   93   92   55    1
```

The codebook says 2 is married, 3 widowed, 4 never married, 5 divorced, 6 separated, and 8 unknown. Two cleanup decisions would be the first to consider. If we do not expect separated and divorced individuals to differ on the outcome, we can pool them, and the single unknown individual is best merged into the largest category rather than giving them a parameter of their own:

```

nhefs$marital <- ifelse(nhefs$marital == 6, 5, Ehefs$marital)
nhefs$marital <- ifelse(nhefs$marital == 8, 2, Ehefs$marital)

table(nhefs$marital)

```

```

##
##      2      3      4      5
## 1238    88    93   147

```

The remaining consideration is the variable's class. `marital` is numeric, and a numeric variable entered as is gets one coefficient, which asserts that the outcome changes linearly across the arbitrary codebook values 2, 3, 4, 5.

For labels with no ordering, that assertion is meaningless. Compare the two codings in a model for smoking intensity:

```

mod_num <- glm(smokeintensity ~ sex + age + marital,
               data = Ehefs, family = poisson("log"))
round(summary(mod_num)$coefficients[, 1:2], 4)

```

```

##              Estimate Std. Error
## (Intercept)    3.3005     0.0265
## sex           -0.2619     0.0113
## age           -0.0028     0.0005
## marital       -0.0122     0.0060

```

```

mod_fac <- glm(smokeintensity ~ sex + age + factor(marital),
               data = Ehefs, family = poisson("log"))
round(summary(mod_fac)$coefficients[, 1:2], 4)

```

```

##              Estimate Std. Error
## (Intercept)    3.2716     0.0228
## sex           -0.2604     0.0115
## age           -0.0026     0.0005
## factor(marital)3 -0.0770     0.0276
## factor(marital)4 -0.1075     0.0247
## factor(marital)5  0.0035     0.0196

```

The numeric coding produced a single uninterpretable slope across the marital status category codes. The factor coding produced a contrast for each category against the married referent, and the categories plainly do not sit on a line: widowed and never married individuals smoke less than married ones, divorced or separated individuals about the same.

The same logic extends to continuous nuisance variables. Age entered linearly is a bet that the confounder outcome relationship is a straight line on the model's scale; if it is not, the unabsorbed structure remains in the model as residual confounding in the target. We can loosen the linearity assumption with a flexible age term:

```
round(summary(glm(smokeintensity ~ sex + age + factor(marital),
  data = nhfs, family = poisson("log")))$coefficients["sex", 1:2], 4)
```

```
## Estimate Std. Error
## -0.2604      0.0115
```

```
round(summary(glm(smokeintensity ~ sex + ns(age, df = 4) + factor(marital),
  data = nhfs, family = poisson("log")))$coefficients["sex", 1:2], 4)
```

```
## Estimate Std. Error
## -0.2658      0.0115
```

Here the sex coefficient barely moves, from  $-0.260$  to  $-0.266$ , so the linearity was adequate in these data.<sup>4</sup> It will not always be: with strong confounding and a curved confounder outcome relationship, the flexibility of the nuisance function is a bias control.

<sup>4</sup> We will soon see the concept of the bias-variance tradeoff, which is something that should govern the decision here about whether we introduce age as a linear term or as a more flexible term.

## 7 Link Functions

The link function  $g$  connects the conditional mean to the linear predictor. Choosing a link function can depend on a few different sets of considerations, but often (particularly when fitting conditionally adjusted regression models) we choose a link function to return a point estimate on a specified scale of interest. For example, the difference scale, the ratio scale, or the odds ratio scale. For this reason, three link functions are most commonly encountered in

the work we do. For a binary outcome with  $P(Y = 1 | X)$  as the conditional mean, we can have:

- the **identity** link, which models  $P(Y = 1 | X)$  itself, so coefficients are risk differences. These models are often referred to as linear risk models.
- the **log** link models  $\log P(Y = 1 | X)$ , so exponentiated coefficients are risk ratios. These are log-linear risk models.
- the **logit** link models  $\log[P(Y = 1 | X) / \{1 - P(Y = 1 | X)\}]$ , so exponentiated coefficients are odds ratios. These are logistic models.

It's essential to understand **why** these links return coefficients on a different scale, and this has entirely to do with the properties of logarithms. With a single binary  $X$ :

$$\begin{aligned}\log[P(Y = 1 | X)] &= \beta_0 + \beta_1 X \\ \implies \beta_1 &= \log[P(Y = 1 | X = 1)] - \log[P(Y = 1 | X = 0)] \\ \implies \beta_1 &= \log\left[\frac{P(Y = 1 | X = 1)}{P(Y = 1 | X = 0)}\right] \\ \implies \exp(\beta_1) &= \frac{P(Y = 1 | X = 1)}{P(Y = 1 | X = 0)}\end{aligned}$$

The same quotient rule reasoning applied to the logit link produces the odds ratio, and with the identity link there is no logarithm to undo, so the coefficient is the risk difference as it stands.

Here are all three links fit to one outcome, an indicator of weight change greater than the 2.6 kg median, with quitting smoking as the exposure:

```
nhefs <- nehs %>%
  mutate(wt_delta = as.numeric(wt82_71 > median(wt82_71)))

round(coef(glm(wt_delta ~ qsmk, data = nehs,
               family = binomial("identity"))), 4)
```

```
## (Intercept)      qsmk
##      0.4669      0.1286
```

```
round(coef(glm(wt_delta ~ qsmk, data = nhfs,
               family = binomial("log"))), 4)
```

```
## (Intercept)      qsmk
##      -0.7616      0.2434
```

```
round(coef(glm(wt_delta ~ qsmk, data = nhfs,
               family = binomial("logit"))), 4)
```

```
## (Intercept)      qsmk
##      -0.1326      0.5195
```

These models yield a risk difference of 0.129, a risk ratio of  $\exp(0.2434) = 1.28$ , an odds ratio of  $\exp(0.5195) = 1.68$ . Arguably, none of these is more “correct” than the others. There is a case to be made that the logistic model is more correct, in a specific sense: this model does not allow predictions from the model to lie outside of  $[0, 1]$ , and as such is a more “correct” model. Note that the linear risk model does not bound the outcome at all, so predictions can effectively lie on  $[-\infty, \infty]$ . The log-linear risk model bounds predictions to lie between  $[0, \infty]$ . Neither of these latter domains is compatible with a probability, and so in this sense logistic regression is more “correct”.

But adopting this position is not always necessary. If we argue that we are not interested in outcome model predictions, but just interested in modeling the mean of an outcome as a function of covariates on different scales, a good case can be made for using link functions in this way. And because the mean of a binary outcome variable is a proportion/probability, using different link functions for binary outcome data is not uncommon.

When we get introduced to generalized linear models, we’ll see the implications of this perspective play out. Specifically, a common misconception holds that using GLMs correctly means choosing the distribution that matches the outcome and then the canonical link that matches the distribution, so a binary outcome “must” get logit. The pairing is convenient, but it is not obligatory. It effectively privileges some abstract mathematical notion of “compatibility” or “congeniality” against our more practical concepts of scientific or substantive relevance and interest.

Another consideration that has more practical consequences is that, because the identity and log links do not bound outcome model predictions to lie on the probability scale, binomial models with these links can (and, in some cases, often do) return fitted values above 1, which can lead to convergence failures (Zou, 2004). The failures and their workarounds are a practical centerpiece of our lectures on GLMs.

## 8 Distributions

The distribution component of a model for how  $Y$  varies around its conditional mean becomes relevant for specific types of models, such as generalized linear models. Specifying a distribution allows us to construct a likelihood function, which enables us to use maximum likelihood estimation as our optimizer. Specifying a distribution implies a specific variance formula (e.g., if we select a binomial distribution for our outcome, this means the variance of our outcome is defined as  $p(1 - p)$ , which is the variance of the binomial distribution), and defines how far observations should scatter from their mean. The variance implications are compactly summarized by each family's mean variance relationship:

| Family   | Conditional mean | Implied variance      |
|----------|------------------|-----------------------|
| Gaussian | $\mu$            | $\sigma^2$ (constant) |
| Binomial | $\mu$            | $\mu(1 - \mu)$        |
| Poisson  | $\mu$            | $\mu$                 |

**Mean variance relationships for three common families.** The Gaussian permits scatter unrelated to the mean; the binomial and Poisson tie the scatter to the mean itself, leaving no free variance parameter.

There are situations where choosing a distribution that functionally ties the variance to the mean (e.g., binomial, or Poisson) can lead to obvious problems. For instance, often times we encounter count outcomes (i.e.,  $Y \in [0, 1, 2, 3, \dots]$ ) whose variance exceeds the mean. This is called overdispersion, and it violates the assumptions embedded in a Poisson model, even when the mean model is right. The usual symptoms are standard errors that are too small, but the coefficients themselves can remain sensible.

This has implications that can be confusing to understand, and relate to the misconception that to get the model “right”, we have to choose the right distribution. Over the course of the next several weeks, we’ll encounter a number of strategies (e.g., such as robust, or “sandwich” variance estimators, or by incorporating overdispersion parameters in the model) that have been developed to loosen our dependence on distributional assumptions for a generalized linear model.

The distribution and the link are separate dials. Of course, if we elect to use the log or logistic link, the scale of the outcome has to be compatible (e.g., we can’t take the log of negative numbers). But other than that, little else prevents us from using a (e.g.) Gaussian family with a log link, as we fit in the left hand side section, or a binomial family with an identity link, as we fit in the link section. Certain pairings are canonical, meaning the mathematics of the family singles out one link as special, Week 6 covers why this is the case.

## 9 Offsets

The offset is less commonly taught, but it is one of the simplest terms in the model: a variable in the right hand side of the model whose coefficient is set to one, and not estimated. We can write an offset as:

$$g[E(Y | X, T)] = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + g[T]$$

which clarifies why it’s called an offset. To get a sense of what this does, consider that weight change between 1982 and 1971 is essentially weight in 1982 minus weight in 1971. If we fit a linear model, we can do this mathematically:

$$\begin{aligned} E(Y | X, T) &= X\beta + T \\ E(Y | X, T) - T &= X\beta \\ E(Y | X, T) - E(T | X, T) &= X\beta \\ E(Y - T | X, T) &= X\beta \end{aligned}$$

Which means that, *for a linear model*, regressing the difference between  $Y$  and  $T$  against  $X$  is the same as regressing  $Y$  against  $X$  with  $T$  as an offset. We can verify this in the NHEFS data:

```
summary(lm(wt82_71 ~ qsmk + age + I(age^2), data = nhefs))$coefficients
```

```
##              Estimate Std. Error  t value    Pr(>|t|)
## (Intercept) -4.859656943  2.489035803 -1.952425 5.106613e-02
## qsmk         3.134849093  0.435903553  7.191612 9.892990e-13
## age          0.496777570  0.114885896  4.324095 1.628497e-05
## I(age^2)     -0.007316666  0.001261203 -5.801339 7.949443e-09
```

```
summary(lm(wt82 ~ qsmk + age + I(age^2), data = nhefs))$coefficients
```

```
##              Estimate Std. Error  t value    Pr(>|t|)
## (Intercept) 53.46722208  5.26672970 10.151883 1.710337e-23
## qsmk         5.15462685  0.92235965  5.588522 2.697799e-08
## age          1.04808537  0.24309532  4.311417 1.723576e-05
## I(age^2)     -0.01321259  0.00266867 -4.951003 8.182899e-07
```

```
summary(lm(wt82 ~ qsmk + age + I(age^2) + offset(wt71), data = nhefs))$coefficients
```

```
##              Estimate Std. Error  t value    Pr(>|t|)
## (Intercept) -4.859656943  2.489035803 -1.952425 5.106613e-02
## qsmk         3.134849093  0.435903553  7.191612 9.892990e-13
## age          0.496777570  0.114885896  4.324095 1.628497e-05
## I(age^2)     -0.007316666  0.001261203 -5.801339 7.949443e-09
```

Note that this does not work if we use a different kind of model, like log-linear or logistic. Seeing why takes us to the offset's canonical use: on the log scale, moving  $g[T] = \log T$  to the left hand side divides the conditional mean by  $T$  rather than subtracting from it, precisely because of the properties of logs:

$$\log Y - \log T = \log \frac{Y}{T}$$

We can use this to construct rate models, which is where the log link count model with  $\log T_i$  becomes a use to us. Let's use an example dataset from the book by [Faraway \(2016\)](#) (page 90). We can obtain the data as follow:

```
pacman::p_load(faraway)

data(dicentric, package = "faraway")

head(dicentric)
```

```
##   cells  ca doseamt doserate
## 1   478  25      1    0.10
## 2  1907 102      1    0.25
## 3  2258 149      1    0.50
## 4  2329 160      1    1.00
## 5  1238  75      1    1.50
## 6  1491 100      1    2.00
```

These data are from a 1976 study of the effect of gamma radiation on the number of chromosomal abnormalities (*ca*) in cells. There were hundreds of cells (*cells*) exposed in each run. The dose amount (*doseamt*) and the dose rate (*doserate*, i.e., the time-span over with the *doseamt* was applied) were measured, and serve as the predictors of interest. We can fit a model with an offset to these data:

```
rate_mod <- glm(ca ~ log(doserate) * factor(doseamt) +
  offset(log(cells)), family = poisson("log"),
  data = dicentric)

summary(rate_mod)$coefficients
```

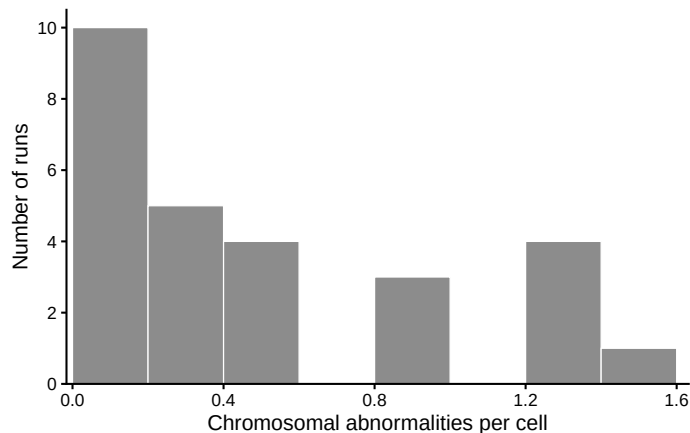
```
##               Estimate Std. Error   z value
## (Intercept)    -2.74671064 0.03426328 -80.164857
## log(doserate)    0.07177751 0.03517620  2.040513
## factor(doseamt)2.5 1.62542047 0.04946041 32.863064
## factor(doseamt)5  2.76108731 0.04348819 63.490512
## log(doserate):factor(doseamt)2.5 0.16122211 0.04830201  3.337793
## log(doserate):factor(doseamt)5  0.19350163 0.04242708  4.560805
##               Pr(>|z|)
```

```
## (Intercept)                0.000000e+00
## log(doserate)              4.129922e-02
## factor(doseamt)2.5         7.412025e-237
## factor(doseamt)5           0.000000e+00
## log(doserate):factor(doseamt)2.5 8.444666e-04
## log(doserate):factor(doseamt)5   5.095789e-06
```

We won't spend too much time reading into these coefficients (there is a strong interaction effect between the log-scale dose rate and dose amount), but the key here is to recognize what the offset in this model is giving us. In effect, we have

$$\begin{aligned}\log E(\text{ca} \mid X, \text{cells}) &= \beta_0 + \beta_1 X + \log \text{cells} \\ \log E(\text{ca} \mid X, \text{cells}) - \log \text{cells} &= \beta_0 + \beta_1 X \\ \log E(\text{ca}/\text{cells} \mid X, \text{cells}) &= \beta_0 + \beta_1 X\end{aligned}$$

So effectively, we have a method to implicitly model the “rate” (here, number of chromosomal abnormalities per cell) as a function of covariates. Why not just model the rate directly? Well, the distribution of  $\text{ca}/\text{cells}$  is as follows:



Note here that there are no zeroes in this distribution:

```
sum((dicentric$ca/dicentric$cells) == 0)
```

```
## [1] 0
```

but the distribution is not an integer count so a Poisson model isn't really compatible. Note the warnings we get, which we limit here to one to avoid pages of warnings:

```
rate_mod_pois <- glm(I(ca/cells) ~ log(doserate) *
  factor(doseamt), family = poisson(link = "log"),
  data = dicentric)
```

```
## Warning in dpois(y, mu, log = TRUE): non-integer x = 0.052301
```

Nevertheless, despite the warnings from the Poisson fit, we still get coefficients:

```
summary(rate_mod_pois)$coefficients
```

| ##                                  | Estimate    | Std. Error | z value     | Pr(> z )   |
|-------------------------------------|-------------|------------|-------------|------------|
| ## (Intercept)                      | -2.76062130 | 1.328703   | -2.07768114 | 0.03773874 |
| ## log(doserate)                    | 0.07438253  | 1.169500   | 0.06360198  | 0.94928715 |
| ## factor(doseamt)2.5               | 1.64901074  | 1.454804   | 1.13349320  | 0.25700719 |
| ## factor(doseamt)5                 | 2.78374862  | 1.371007   | 2.03044030  | 0.04231180 |
| ## log(doserate):factor(doseamt)2.5 | 0.14922629  | 1.291189   | 0.11557274  | 0.90799117 |
| ## log(doserate):factor(doseamt)5   | 0.18321198  | 1.211158   | 0.15127015  | 0.87976262 |

Instead, we could use a gamma GLM with a log link:

```
rate_mod2 <- glm(I(ca/cells) ~ log(doserate) *
  factor(doseamt), family = Gamma(link = "log"),
  data = dicentric)
```

```
summary(rate_mod2)$coefficients
```

| ##                    | Estimate    | Std. Error | t value    | Pr(> t )     |
|-----------------------|-------------|------------|------------|--------------|
| ## (Intercept)        | -2.76068672 | 0.03319989 | -83.153496 | 6.141109e-28 |
| ## log(doserate)      | 0.07542757  | 0.02837912 | 2.657855   | 1.471991e-02 |
| ## factor(doseamt)2.5 | 1.64698321  | 0.04695173 | 35.078221  | 3.975217e-20 |
| ## factor(doseamt)5   | 2.78204793  | 0.04695173 | 59.253362  | 7.344802e-25 |

```
## log(doserate):factor(doseamt)2.5  0.16170915 0.04013414  4.029217 6.059026e-04
## log(doserate):factor(doseamt)5    0.19238406 0.04013414  4.793527 9.773451e-05
```

which, as an aside, does require that there are no zeros in the outcome:

```
dicentric2 <- dicentric |>
  mutate(rate_outcome = if_else((ca/cells) <
    0.2, 0, ca/cells))

rate_mod_test <- glm(rate_outcome ~ log(doserate) *
  factor(doseamt), family = Gamma(link = "log"),
  data = dicentric2)
```

```
## Error:
```

```
## ! non-positive values not allowed for the 'Gamma' family
```

The table below places the estimates and standard errors from `rate_mod`, `rate_mod_pois`, and `rate_mod2` side by side, with the HC3 robust standard error added for the Poisson model of the rate:

| Term                        | Poisson, offset |       | Poisson, rate |       |        | Gamma, rate |       |
|-----------------------------|-----------------|-------|---------------|-------|--------|-------------|-------|
|                             | Estimate        | SE    | Estimate      | SE    | HC3 SE | Estimate    | SE    |
| Intercept                   | -2.747          | 0.034 | -2.761        | 1.329 | 0.024  | -2.761      | 0.033 |
| log(doserate)               | 0.072           | 0.035 | 0.074         | 1.170 | 0.018  | 0.075       | 0.028 |
| doseamt 2.5                 | 1.625           | 0.049 | 1.649         | 1.455 | 0.056  | 1.647       | 0.047 |
| doseamt 5                   | 2.761           | 0.043 | 2.784         | 1.371 | 0.053  | 2.782       | 0.047 |
| log(doserate) : doseamt 2.5 | 0.161           | 0.048 | 0.149         | 1.291 | 0.065  | 0.162       | 0.040 |
| log(doserate) : doseamt 5   | 0.194           | 0.042 | 0.183         | 1.211 | 0.054  | 0.192       | 0.040 |

Provided that there are no zeroes in the outcome (which is incompatible with a gamma distribution). We get more into the distinctions between GLMs with different distributions next week. For now, note how we can, in this example, use a Poisson or gamma regression model and still get results that roughly align (at least in terms of their point estimates).

## 10 The Parametric-to-Nonparametric Spectrum

We've seen the pieces that can go into a regression model. And we've seen in a few instances last week how a model is essentially a "lossy compression"

device. It takes the data as input, and compresses it to represent only the features we wish to emphasize. In the kidney example, you saw that one of the features we focused on was the average difference between kidney lengths among infected and non-infected children. But in doing so, we assumed that this difference was the same across all ages.

This idea is connected to the concept of non-parametric, semiparametric, and parametric models. Let's illustrate this idea in a bit more depth. We start with a simple simulated example with three variables  $x_1$ ,  $x_2$ , and  $y$ , all binary. These data can be summarized (without any loss of information) in a stratified two-by-two table:

|                             | $y = 1$ | $y = 0$ |
|-----------------------------|---------|---------|
| <b><math>x_2 = 0</math></b> |         |         |
| $x_1 = 1$                   | 62      | 200     |
| $x_1 = 0$                   | 23      | 218     |
| <b><math>x_2 = 1</math></b> |         |         |
| $x_1 = 1$                   | 115     | 116     |
| $x_1 = 0$                   | 53      | 213     |

These data, as an object in R, look like this:

```
head(dat_)
```

```
##   y x1 x2
## 1 0  0  0
## 2 1  1  1
## 3 0  0  0
## 4 1  1  1
## 5 1  1  1
## 6 0  0  0
```

In such a setting with a binary outcome, and two binary covariates, we can fit a logistic regression model that regresses the outcome against the two covariates, and the interaction between the two:

$$\text{logit } P(Y = 1 \mid X, C) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2$$

The combination of  $X_1$  and  $X_2$  gives us four possible levels in total. This means that each sub-group of the data defined by the combination of  $X_1$  and

$X_2$  can potentially have its own unique outcome probability. Note also how the logistic model contains four parameters, one for each  $X_1$  and  $X_2$  combination.

Because of this, we say the model is “saturated” (i.e., we can’t add any more parameters to this model, even if we wanted too). And because we are using maximum likelihood here to estimate logistic model parameters, we have what’s called nonparametric maximum likelihood estimator.

We can implement this in our data easily:

```
mod1 <- glm(y ~ x1 * x2, data = dat_, family = binomial("logit"))

summary(mod1)$coefficients
```

| ## |             | Estimate   | Std. Error | z value     | Pr(> z )     |
|----|-------------|------------|------------|-------------|--------------|
| ## | (Intercept) | -2.2490008 | 0.2192095  | -10.2595963 | 1.071538e-24 |
| ## | x1          | 1.0778178  | 0.2630243  | 4.0977876   | 4.171177e-05 |
| ## | x2          | 0.8580006  | 0.2676108  | 3.2061506   | 1.345236e-03 |
| ## | x1:x2       | 0.3045244  | 0.3317543  | 0.9179215   | 3.586600e-01 |

We can make this model unsaturated, or less flexible, by removing the interaction between the covariates:

```
mod2 <- glm(y ~ x1 + x2, data = dat_, family = binomial("logit"))

summary(mod2)$coefficients
```

| ## |             | Estimate  | Std. Error | z value    | Pr(> z )     |
|----|-------------|-----------|------------|------------|--------------|
| ## | (Intercept) | -2.387082 | 0.1688116  | -14.140512 | 2.137252e-45 |
| ## | x1          | 1.272461  | 0.1613131  | 7.888143   | 3.067153e-15 |
| ## | x2          | 1.058907  | 0.1591397  | 6.653944   | 2.853404e-11 |

but because it’s not saturated, it’s no longer being estimated with nonparametric maximum likelihood estimation.

We can see what this means by comparing the fitted probabilities from the two models in each of the four covariate patterns:

|               |     | P(y = 1) |           |                |
|---------------|-----|----------|-----------|----------------|
|               | n   | Observed | Saturated | No interaction |
| <b>x2 = 0</b> |     |          |           |                |
| x1 = 1        | 262 | 0.237    | 0.237     | 0.247          |
| x1 = 0        | 241 | 0.095    | 0.095     | 0.084          |
| <b>x2 = 1</b> |     |          |           |                |
| x1 = 1        | 231 | 0.498    | 0.498     | 0.486          |
| x1 = 0        | 266 | 0.199    | 0.199     | 0.209          |

The saturated model returns the observed proportion in each cell exactly: with four parameters for four cells, there is nothing to smooth. The model without the interaction has three parameters for the same four cells, so it cannot match all four proportions and instead smooths across them, pulling two cells up and two cells down. Exactly **how** the model pulls the cells up and down is a little involved, but has been the topic of research dating back to at least the early 1900s ([Bartlett, 1935](#)). Suffice it to say that there is no simple closed form solution telling us exactly how this occurs. One way to think about it will be through the framework of iteratively re-weighted least squares, which we cover next week.



**Context Note:** The Technical Definitions of Nonparametric, Semiparametric, and Parametric Models

A statistical model is the set of probability distributions we are willing to consider for the data at hand, written  $\mathcal{M} = \{P_\theta : \theta \in \Theta\}$ , where the index  $\theta$  is called the parameter and the set  $\Theta$  it lives in is the parameter space. The three labels in this section are statements about the dimension of  $\Theta$  (Bickel et al., 1993; van der Vaart, 2000). A **nonparametric** model imposes essentially no restriction:  $\Theta$  is the collection of all distributions (compatible with basic regularity conditions, for instance, finite means, or differentiable loss), and because no finite list of numbers can index every candidate distribution, the parameter is infinite dimensional, a function rather than a vector, like the unrestricted  $f$  in  $E(Y | X) = f(X)$ . This is the technical sense in which the saturated model above is nonparametric: with two binary covariates, “all possible conditional distributions” happens to be exhausted by four probabilities, so the four coefficient logistic model imposes no restriction at all, and its maximum likelihood estimator is the NPMLE.

A **semiparametric** model splits the parameter into two parts  $\theta = (\beta, \eta)$ : a finite dimensional part  $\beta$  that is not nonparametric (because it is finite dimensional), and an infinite dimensional piece  $\eta$  that is left as an unrestricted function (Tsiatis, 2006; Kennedy, 2016). One way to think of this is in “target/nuisance” function terms, but this need not be the case. For instance, we can specify precisely the component we care about, but refuse to put a functional form on the rest. The flagship example of this is the Cox model, where the association with covariates is captured by a finite  $\beta$  while the baseline hazard, an entire function of time, is never specified.

A **parametric** model is one where  $\Theta$  is a subset of  $\mathbb{R}^k$  for some finite  $k$ : a list of  $k$  numbers describes everything the model permits about the distribution of the data. The unsaturated logistic model above is the smallest possible example of the move: dropping the product term leaves three parameters to describe four covariate patterns, and the model now asserts a restriction (no interaction on the log odds scale) that the data are not permitted to contradict. When a restriction like this is right, it buys precision; when it is wrong, we have model misspecification.

We can see this play out in a still simple, but slightly more complicated dataset like `kidneys`. Here, we focus on the relationship between just kidney length and age (i.e., no infection), and plot three relationships: linear, curvilinear, and saturated.

To fit the curvilinear model, we use squared and cubic terms for age, but could just have easily used fractional polynomials or splines.

To fit the saturated relationship, we included age as a “dummy” variable using the `factor()` function in R. This gives us a total of 176 unique parameters that go into the regression model.

```

Kidneys <- read_csv(here("data", "kidneys.csv"), skip = 1)

kidney_model1 <- lm(length ~ age,
                    data = kidneys)

kidney_model2 <- lm(length ~ age + I(age^2) + I(age^3),
                    data = kidneys)

kidney_model3 <- lm(length ~ factor(age),
                    data = kidneys)

c(length(coef(kidney_model1)), length(coef(kidney_model2)),
  length(coef(kidney_model3)))

## [1] 2 4 176

```

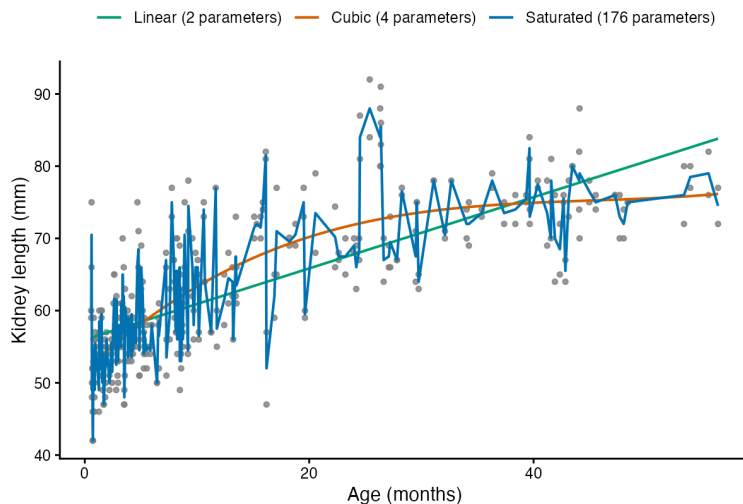


Figure 2: Kidney length against age, fit three ways: 350 kidney measurements from 180 children (most contribute both kidneys). The linear model (2 parameters) tracks the overall trend, and the cubic (4 parameters) bends with early growth. The saturated model (176 parameters, one mean per distinct age) reproduces each age's mean exactly and imposes no structure between ages. Where two measurements share an age, its fit passes through their average rather than through each point, which is why even 176 parameters do not trace every observation.

Consider a final simple(-ish) setting, but this time with two continuous variables instead of binary ones:

```

set.seed(123)

n = 250

```

```
## Note here: we round to three decimal
## places, and use the 'factor()'
## function This gives us dummy
## variable coding for x1 and x2
x1 <- factor(round(rnorm(n), 3))
x2 <- factor(round(rnorm(n), 3))

dat_ <- data.frame(x1, x2)

head(dat_, 10)
```

```
##      x1      x2
## 1  -0.56 -0.376
## 2  -0.23 -0.562
## 3   1.559 -0.344
## 4   0.071  0.09
## 5   0.129  1.599
## 6   1.715 -0.089
## 7   0.461  1.081
## 8  -1.265  0.631
## 9  -0.687 -0.114
## 10 -0.446 -1.533
```

```
mod_mat <- model.matrix(~., data = dat_)

dim(mod_mat)
```

```
## [1] 250 480
```

```
mod_mat_int <- model.matrix(~.^2, data = dat_)

dim(mod_mat_int)
```

```
## [1] 250 57840
```

Even if we (1) round all continuous variables to three decimal places, and (2) ignore any potential interactions, and include a parameter for every level of both variables in the model, we end up with a total of 480 parameters in the model. Adding two way interactions alone increases this number to 57840 parameters. Consequently, in any realistic setting, the NPMLE will be undefined, and it will not be possible to use it for quantifying a regression model.

In this illustrative example, the main takeaway should be that nonparametric methods impose little to no structure about the underlying shapes in the data. To use Sander's turn of phrase, it is a "lossless compression" as opposed to a "lossy compression". This is great, and so we should use nonparametric all the time. But the problem is that in most realistic settings, we can't because there's simply too much information in the data to "compress" into a model without loss.

So we opt for semiparametric methods when we can, or flexible parametric methods when we can't.

## 11 Conclusion

| Components      | What we choose                                         | What the choice decides                        |
|-----------------|--------------------------------------------------------|------------------------------------------------|
| Left hand side  | The outcome, its coding, and central tendency function | The summary of the outcome                     |
| Right hand side | Variables, coding, flexibility, interactions, form     | How flexible the model is, what adjusted means |
| Link            | Scale of additivity (linear, log-linear, logistic)     | Contrast type: difference, ratio, odds ratio   |
| Distribution    | Variation around the mean                              | The likelihood; the standard errors            |
| Offset          | Terms with known coefficients                          | Rates, standardization, external knowledge     |

This week broke the regression machinery apart, and we sought to demonstrate what these pieces are and what they do. This leads to the idea that a regression model is a stack of separable choices: an outcome coding and central tendency function (mean, quantile) that selects what we are regressing, a right hand side that defines the functional form and, when relevant, splits into a target function coded for interpretation and a nuisance function coded for performance, a link that selects the contrast scale, a distribution that stands behind the likelihood and the standard errors, and an offset that carries what is

already known.

None of these choices are contained in the data. Most have defaults that may or may not be reasonable for the questions you are asking.

## References

- Douglas G Altman and Patrick Royston. The cost of dichotomising continuous variables. *BMJ*, 332(7549):1080, May 2006.
- Joshua D. Angrist and Jörn-Steffen Pischke. *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton University Press, January 2009. ISBN 978-0-691-12035-5. Google-Books-ID: YSAzEAAAQBAJ.
- M. S. Bartlett. Contingency table interactions. *Supplement to the Journal of the Royal Statistical Society*, 2(2):248–252, 1935. DOI: 10.2307/2983639.
- Caroline Bennette and Andrew Vickers. Against quantiles: categorization of continuous variables in epidemiologic research, and its discontents. *BMC Medical Research Methodology*, 12:21, 2012.
- P. J. Bickel, C. A. J. Klaassen, Y Ritov, and J. A. Wellner. *Efficient and adaptive estimation for semiparametric models*. Springer-Verlag, New York, 1993.
- John B Carlin and Margarita Moreno-Betancur. On the uses and abuses of regression models: A call for reform of statistical practice and teaching. *Statistics in Medicine*, 44(13-14):e10244, 2025.
- Victor Chernozhukov, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, and James Robins. Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1):C1–C68, 2018.
- Stephen Colagiuri, Crystal M. Y. Lee, Tien Y. Wong, Beverley Balkau, Jonathan E. Shaw, Knut Borch-Johnsen, and DETECT-2 Collaboration Writing Group. Glycemic thresholds for diabetes-specific retinopathy: Implications for diagnostic criteria for diabetes. *Diabetes Care*, 34(1):145–150, 2011. DOI: 10.2337/dc10-1206.
- Jamie DeCoster, Anne-Marie R. Iselin, and Marcello Gallucci. A conceptual and empirical examination of justifications for dichotomization. *Psychological Methods*, 14(4):349–366, 2009. DOI: 10.1037/a0016956.
- Julian J. Faraway. *Extending the Linear Model with R: Generalized Linear, Mixed Effects and Nonparametric Regression Models, Second Edition*. Chapman and

Hall/CRC, New York, 2 edition, March 2016. ISBN 978-1-315-38272-2. DOI: 10.1201/9781315382722.

David P. Farrington and Rolf Loeber. Some benefits of dichotomization in psychiatric and criminological research. *Criminal Behaviour and Mental Health*, 10(2):100–122, 2000. DOI: 10.1002/cbm.349.

Ragnar Frisch and Frederick V. Waugh. Partial time regressions as compared with individual trends. *Econometrica*, 1(4):387–401, 1933.

Sander Greenland, Malcolm Maclure, James J. Schlesselman, Charles Poole, and Hal Morgenstern. Standardized regression coefficients: A further critique and review of some alternatives. *Epidemiology*, 2(5):387–392, 1991.

M. A. Hernán and JM Robins. *Causal Inference*. Chapman & Hall/CRC, Boca Raton, FL, 2020.

Edward H Kennedy. Semiparametric theory and empirical processes in causal inference. In Hua He, Pan Wu, and Ding-Geng (Din) Chen, editors, *Statistical Causal Inferences and Their Applications in Public Health Research*. Springer International, Switzerland, 2016.

Roger Koenker and Jr. Bassett, Gilbert. Regression quantiles. *Econometrica*, 46(1):33–50, 01 1978.

Michael C. Lovell. Seasonal adjustment of economic time series and multiple regression analysis. *Journal of the American Statistical Association*, 58(304): 993–1010, 1963.

Boyd E. Metzger, Lynn P. Lowe, Alan R. Dyer, Elisabeth R. Trimble, Udom Chao-varindr, Donald R. Coustan, David R. Hadden, David R. McCance, Moshe Hod, H. David McIntyre, Jeremy J. N. Oats, Bengt Persson, Michael S. Rogers, David A. Sacks, and HAPO Study Cooperative Research Group. Hyperglycemia and adverse pregnancy outcomes. *New England Journal of Medicine*, 358(19):1991–2002, 2008. DOI: 10.1056/NEJMoa0707943.

Boyd E. Metzger, Steven G. Gabbe, Bengt Persson, Thomas A. Buchanan, Patrick A. Catalano, Peter Damm, Alan R. Dyer, Alberto de Leiva, Moshe Hod, John L. Kitzmiller, Lynn P. Lowe, H. David McIntyre, Jeremy J. N. Oats, Yasue Omori, and Maria Ines Schmidt. International association of diabetes

- and pregnancy study groups recommendations on the diagnosis and classification of hyperglycemia in pregnancy. *Diabetes Care*, 33(3):676–682, 2010. doi: 10.2337/dc09-1848. International Association of Diabetes and Pregnancy Study Groups Consensus Panel.
- Ashley I. Naimi, Qianhui Jin, Ya-Hui Yu, Sara M. Parisi, and Lisa M. Bodnar. Residual-on-Residual Regression as a Tool for Effect Estimation in Observational Data, June 2026.
- P. M. Robinson. Root-N-Consistent Semiparametric Regression. *Econometrica*, 56(4):931–954, 1988. ISSN 0012-9682. doi: 10.2307/1912705.
- Jasper M. Schellingerhout, Martijn W. Heymans, Henrica C. W. de Vet, Bart W. Koes, and Arianne P. Verhagen. Categorizing continuous variables resulted in different predictors in a prognostic model for nonspecific neck pain. *Journal of Clinical Epidemiology*, 62(8):868–874, 2009.
- Anastasios A. Tsiatis. *Semiparametric theory and missing data*. Springer, New York, 2006.
- A. W. van der Vaart. *Asymptotic statistics*. Cambridge University Press, Cambridge, 2000.
- Daniel Westreich and Sander Greenland. The table 2 fallacy: Presenting and interpreting confounder and modifier coefficients. *American Journal of Epidemiology*, 177(4):292–298, 2013.
- Guangyong Zou. A modified poisson regression approach to prospective studies with binary data. *Am J Epidemiol*, 159(7):702–706, Apr 2004.